

RESEARCH STATEMENT

Paolo Bordinon

p.bordinon@math.leidenuniv.nl

paolobordinon.github.io

github.com/paolobordinon

OVERVIEW

Throughout my mathematical research, I have always been fascinated and pulled by **arithmetic phenomena** firstly observed **analytically** – values of automorphic functions, meromorphic modular forms, Laplacian spectral patterns – that seem to have a deep **geometric nature** involving moduli spaces and motives. I am drawn to the **p -adic world** because it is precisely there that the two can be reconciled: a class in p -adic cohomology is at once an analytic object, a differential form or a formal Fourier expansion, and a geometric one, carrying a Frobenius action, a filtration and a Galois representation. In my thesis work I have studied in depth p -adic aspects of modular curves; the main project involves filtered (φ, N) -modules and Nekovář’s height pairings [Nek93] to construct explicit p -adic analogues of the **higher Green’s functions** §1.1 of Gross–Kohnen–Zagier [GKZ87] on Shimura curves. In the same area, I have proven the conjectures of [Zha25] explaining congruences between the **Fourier coefficients of meromorphic modular forms** §1.2 and the Frobenius eigenvalues of the elliptic curves along which they have poles. The spectral properties of the Laplacian seem to have remarkable similarities with the **spectral theory of the U_p -operator** §1.3, where the analytic behaviour of overconvergent eigenforms is dictated by the geometry of the supersingular locus and by deformations of p -divisible groups.

1 RESEARCH WORK: THREE p -ADIC APPROACHES TO ANALYTIC PHENOMENA

1.1 p -adic heights on Shimura curves and higher p -adic Green functions (joint with Hazem Hassan, available on request)

The seminal work of Gross–Zagier on the BSD conjecture [GZ86] and the following work [GKZ87] have shed light on interesting connections between automorphic functions, intersection theory and modular objects in general. The higher Green functions $G_k(z, w)$ are one of those cornerstone objects in the proof and subsequent works since they connect the **hyperbolic Laplace operator** to **intersection theory of Heegner cycles**. The value of these functions at CM points has been conjectured, in the same paper, to be equal to the **logarithm of an algebraic number**. This has sparked research in several directions and has now recently been fully resolved in many instances by fairly different methods. The nature of the object has indeed attracted researchers both from a more analytic side [BEY21], [Via11] and [Li23] via the theory of Borchers lifts and analytic Maass forms and from a geometric one [Zha97], [Mel08] and [BF25] using a more motivic approach. This work reveals that G_k is best understood as the **real part of a period of a specific extension of mixed motives**, computed via an explicit generating-series primitive. This gave us a natural template for a p -adic construction: if higher Green’s functions are periods of extensions in the archimedean setting, their p -adic analogues should be **p -adic heights of the corresponding extensions** of Galois representations, in the sense of Nekovář [Nek93] and Coleman–Gross.

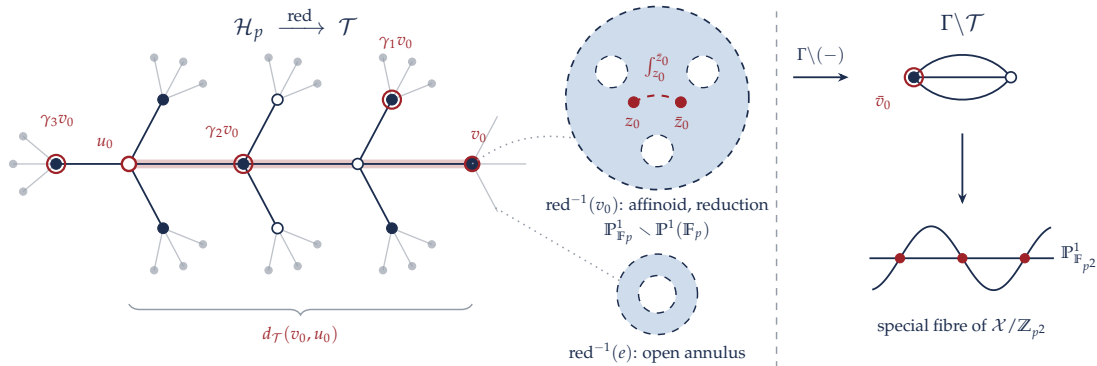


Figure 1: The Čerednik–Drinfeld uniformisation at a prime p . *Left*. Drinfeld’s upper half plane $\mathcal{H}_p = \mathbb{P}^1_{\mathbb{C}_p} \setminus \mathbb{P}^1(\mathbb{Q}_p)$ retracts onto the $(p+1)$ -regular Bruhat–Tits tree $\mathcal{T}$ of $\mathrm{PGL}_2(\mathbb{Q}_p)$, whose vertices are the homothety classes of $\mathbb{Z}_p$ -lattices in $\mathbb{Q}_p^2$. The fibre of red over a vertex is an affinoid with canonical reduction $\mathbb{P}^1_{\mathbb{F}_p} \setminus \mathbb{P}^1(\mathbb{F}_p)$ while over an open edge is an open annulus. *Right*. $X_{\mathbb{Q}_{p^2}} \cong \Gamma \backslash \mathcal{H}_p$. The finite graph $\Gamma \backslash \mathcal{T}$ is the dual graph of the special fibre, whose components are $\mathbb{P}^1_{\mathbb{F}_{p^2}}$ meeting at ordinary double points.

In joint work with Hazem Hassan, I have developed a p -adic analogue of the **higher Green's functions** $G_k(z, w)$, and realized them as explicit local p -adic heights of CM cycles on Shimura curves. Working on Shimura curves uniformized p -adically via Čerednik–Drinfeld (Figure 1) we constructed extensions supported at CM points, where p is inert, producing **biextensions of semistable Galois representations** to which Nekovář's general local height pairing formalism applies. The heart of the project is showing that this abstract height admits a completely explicit formula. We construct the Frobenius- and monodromy-equivariant splitting needed to evaluate the pairing via a renormalized **Coleman integration**. In the paper we have extended Besser's [Bes02] and Besser–Zerbes's [Bes22] [BZ23] p -adic height approach from trivial to arbitrary symmetric-power coefficients, and we adapt Brown–Fonseca's generating-series computation of the archimedean primitive to the p -adic, rigid-analytic setting. The resulting formula expresses the p -adic height first as the Coleman integral of a Poincaré series and then as a sum over the uniformizing arithmetic group Γ

$$G_k(z_0, w_0) = \int_{z_0}^{\bar{z}_0} \Psi_k(z, w_0, \bar{w}_0) dz = \sum_{\gamma \in \Gamma} Q_m(1 - C(\gamma z_0, w_0)). \quad (1)$$

with Q_m Legendre function of the second kind, C cross ratio.

This is a direct structural analogue of the classical Gross–Kohnen–Zagier formula, and, in the base case of weight two, a generalization compatible with the classical Coleman–Gross p -adic height of Heegner points. This approach gives a blueprint to prove the p -adic log algebraicity of the p -adic higher Green functions following [BF25]. Using the fundamental result of **Fontaine's semistable conjecture**, proved by Tsuji [Tsu99] and Faltings [Fal02], we can act by the Hecke algebra on the global cohomology to make the biextension a simple extension of Tate motives, i.e. a **Kummer motive**. This part of the work is currently under development.

1.2 A p -adic cohomological approach to congruences of meromorphic modular forms (arXiv)

In order to explain the content of the paper and the main result we can start with the following phenomenon first observed and conjectured by Pengchang Zhang (MPIM, IHES) in [Zha25]. Let $C/\mathbb{Q}$ be the elliptic curve defined by the equation

$$C/\mathbb{Q} : y^2 + xy = x^3 - x^2 - 2x - 1. \quad (2)$$

For every prime p , an important arithmetic invariant associated to the curve is the number of points defined over the finite field $\mathbb{F}_p$. Define the usual quantity $a_p(C) = p + 1 - \#C(\mathbb{F}_p)$. Consider the following meromorphic function defined on the complex upper half plane $\mathbb{H} = \{\tau \in \mathbb{C} : \text{Im}\tau > 0\}$

$$f(\tau) = \frac{E_4(\tau)}{j(\tau) + 3375} \quad (3)$$

where for $\tau \in \mathbb{C}$, $\text{Im}(\tau) > 0$ and $q = e^{2\pi i \tau}$, we have E_4 the weight four **Eisenstein series** and the j -function associated to the elliptic curve $C/(\mathbb{Z} + \tau\mathbb{Z})$. Zhang observed systematic congruences linking **Fourier coefficients** of meromorphic functions to roots of the **characteristic polynomial of Frobenius** $X^2 - a_p(C)X + p$. For every ordinary prime $p > 3$ and $\ell \geq 0$, letting $u_p(C)$ be the p -adic unit root of the polynomial:

$$a_{np^{\ell+1}}(f) \equiv u_p(C)^2 \cdot a_{np^\ell}(f) \pmod{p^{3\ell}}.$$

This surprising phenomenon links **analytic data** coming from the meromorphic modular form and **geometric data** of the elliptic curve, in particular it connects the U_p -action on meromorphic modular forms and Frobenius eigenvalues on cohomology of the elliptic curve. In the paper we give a p -adic cohomological framework on the moduli space of elliptic curves to explain the congruences. It is key to consider the **rigid cohomology** on modular curves, the theory of **overconvergent modular forms** and the **p -adic geometry** of the modular curve. Let X be a fine moduli space of elliptic curves with extra structure, $\pi : \mathcal{E} \rightarrow X$ the universal elliptic curve. The vector bundle $\mathcal{H} = R^1\pi^*(\Omega_{\mathcal{E}/X}^\bullet)$ comes endowed with a connection and has the structure of an F -isocrystal. We have a short exact sequence of filtered φ -modules:

$$\begin{array}{ccccccc} 0 & \longrightarrow & H_{\text{dR}}^1(X, \text{Sym}^k \mathcal{H}) & \longrightarrow & H_{\text{dR}}^1(X \setminus \{j(C)\}, \text{Sym}^k \mathcal{H}) & \xrightarrow{\text{Res}} & \bigoplus_{\alpha \in |j(C)|} \text{Sym}^k H_{\text{dR}}^1(\mathcal{E}_\alpha)[1] \longrightarrow 0 \\ & & & & \downarrow \wr & & \\ & & & & M_{k+2}^{\text{mero}, j(C)} / \left(q \frac{d}{dq}\right)^{k+1} M_{-k}^{\text{mero}, j(C)} & & \end{array} \quad (4)$$

As shown in the diagram, the central term is isomorphic to the space of weight $k + 2$ meromorphic modular forms with poles along the elliptic curves with j -invariant $j(C)$ modulo the image of an explicit differential operator. Using an idea of Coleman [Col96] involving Katz-Lubin **canonical subgroup**, we can give an explicit description of the action of Frobenius on the cohomology group in terms of the Hecke endomorphism U_p . Carefully using crystalline cohomology on the formal neighborhood at the cusp $[\infty]$ implies the congruences found by P. Zhang.

The core of the project is translating the initial congruence into a modular machinery where the study of extension of filtered φ -modules by simple pieces ties the analytic and geometric data. As a byproduct of this proof of the conjecture of Zhang, this work shed some light on a modular and explicit action of Frobenius on cohomology that can be relevant for recent work in the **explicit quadratic Chabauty area** for developing algorithms that do not depend on models of the curves [Ren25].

1.3 Spectral aspects of p -adic Maaß forms (joint with Jan Vonk)

A theme running through my thesis is that the analytic Laplacian Δ seems to have a counterpart in p -adic geometry. A striking instance of this, going back to suggestions of Blasius and Calegari, is the analogy

$$\left\{ \begin{array}{l} \text{Maaß forms } \mathcal{M}_{\infty}^{\text{an}} \\ \Delta = -y^2(\partial_x^2 + \partial_y^2) \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{l} \text{Overconvergent forms } \mathcal{M}_p^{\text{an},r} \\ " \Delta " = \log |U_p| \end{array} \right\}$$

between the **hyperbolic Laplacian** acting on **Maaß forms** and the **Atkin operator** U_p acting on the space $M_p^0(r)$ of **r -overconvergent modular functions**, under which the Selberg spectral gap corresponds to the smallest positive slope of U_p , Weyl's law to the asymptotics of the slope sequence [Wan98], and the nodal domains of a Maaß eigenfunction [HR92] to the zero locus of an overconvergent one. Archimedean equidistribution of such zero loci is a theorem for holomorphic forms [Rud05]; on the p -adic side the patterns that have been observed come with no geometric mechanism, and supplying one is the aim of this project.

Let p be a prime for which $X_0(p)$ has genus zero, let $h = (\Delta(p\tau)/\Delta(\tau))^{1/(p-1)}$ be the standard Hauptmodul and $v(A)$ the valuation of a lift of the Hasse invariant. The explicit decompositions of U_p obtained by Buzzard–Calegari [BC05] and Loeffler [Loe07] imply that the eigenfunctions f_n on $M_p^0(r)$, ordered by increasing slope have exactly n zeroes on X_r . We have computed these zero divisors for $p = 2, 3, 5, 7, 13$, and find that their radii not only accumulate at $v(A) = 1/(p+1)$ but appear to obey an exact law: on the component $n \equiv 1 \pmod{p-1}$ we conjecture

$$\text{Div}(f_{p^n}) = [\infty] + \sum_{k=0}^{n-1} p^k(p-1) \text{ roots of radius } r_k, \quad r_k = \frac{1}{p+1} - \frac{1}{\text{lcm}(12, p+1) p^k}.$$

The degree is p^n and the barycentre tends to $1/(p+1)$; what the conjecture adds to a bare equidistribution statement is that the zero divisors themselves converge, f_{p^n} seeing the first n members of one infinite family of radii (Figure 2).

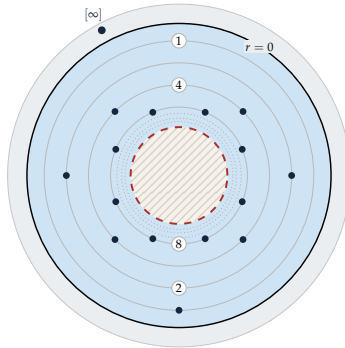


Figure 2: The zero divisor of the p -depleted Eisenstein series $E_0^{[p]} = \log_p(V(h)/h^p)$. The sub-annulus shown is for $p = 2$. The zeroes lie on the circles r_k with multiplicity $p^k(p-1)$, here $k = 0, 1, 2, 3$, accumulating at the boundary, together with a simple zero at the cusp $[\infty]$. For $p = 2$ the radii are $\frac{1}{4}, \frac{7}{24}, \frac{5}{16}, \frac{31}{96}, \dots \rightarrow \frac{1}{3}$; the first n circles form $\text{Div}(f_{2^n})$. A drawn circle is the whole locus $v(A) = r$.

Calegari observed [Cal13] that f_{2^i} converges to the **2-depletion** $E_0^{[2]}$ of the weight zero Eisenstein series; we conjecture in general that

$$\lim_{n \rightarrow \infty} f_{p^n} = \sum_{p \nmid m} \chi^{(p-1)/2}(m) \sigma_{-1}(m) q^m,$$

where χ is the Teichmüller character, so that the limit is the **Eichler integral** of a weight two Eisenstein series with quadratic nebentypus, an infinite slope form of the kind studied in [BGK12]. Since $E_0^{[p]} = \log_p(V(h)/h^p)$, the zeroes of the limit function are the preimages of μ_{p^∞} under $V(h)/h^p$, and the primitive p^{k+1} -th roots of unity number $\varphi(p^{k+1})$ — exactly the multiplicity above.

We evaluate $V(h)/h^p$ on the formal group of the universal elliptic curve using Kato’s theta functions [Kat04] and the Siegel units they produce, together with the canonical subgroup theory of Katz [Kat73] and Lubin [Lub79]. This identifies the accumulation radius as the threshold at which the canonical subgroup of order p^2 ceases to exist. Determining the exact valuation of $V(h)/h^p - 1$ as a function of $v(A)$, which is work in progress, would locate every zero of the limit function.

2 FUTURE DIRECTIONS

The three projects above are connected by more than a shared toolbox. The dichotomy that organises the spectral picture of §1.3 is remarkably similar to the one that separates the archimedean from the non-archimedean local height contributions in [GZ86] and [GKZ87], which are precisely the objects that §1.1 computes p -adically. That one splitting should govern both the analytic behaviour of U_p and the decomposition of a p -adic height pairing is unlikely to be an accident, and making it structural is the direction in which I would most like to push this circle of ideas.

A p -adic Gross–Zagier formula on Shimura curves. The formula of §1.1 evaluates local heights of CM cycles; the natural next step is to relate their global counterpart to derivatives of p -adic L -functions, in the higher weight setting and for the quaternionic curves in the spirit of the archimedean results of [Zha97]. Because the expression we obtained is a Coleman integral of an explicit Poincaré series, a formula of this shape would come with a means of computing the relevant derivatives, and not only with an existence statement.

Spectral asymptotics for the U_p -operator. Beyond the zero divisor conjecture of §1.3, the dictionary with Maaß forms predicts asymptotic statements: a Weyl law for the slope sequence, and equidistribution of the zero loci as the slope grows, the p -adic counterpart of [Rud05]. My conjecture describes exact radii for one family at small genus; the goal is the corresponding asymptotics, uniformly in the prime and the level, where the geometry of the modular curve at infinite level provides tools with no archimedean analogue.

Explicit Frobenius and algorithms. The description of the Frobenius action obtained in §1.2 is explicit enough to compute with, and I intend to develop it into algorithms that require no choice of model for the curve, continuing the line of [Ren25]. This is the most computational strand of my work, and the one in which the code I have written for Frobenius on de Rham cohomology and for Gauss–Manin connections in families should be directly useful.

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